

Лабораторная работа №12

Администрирование локальных сетей

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Информация

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Вводная часть

- Приобретение практических навыков по настройке доступа локальной сети к внешней сети посредством NAT

Основная часть

Настройка маршрутизатора provider-gw-1

- Первоначально настроим маршрутизатор provider-gw-1

```
provider-arromichina-gw-1>enable
provider-arromichina-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
provider-arromichina-gw-1(config)#line vty 0 4
provider-arromichina-gw-1(config-line)#password cisco
provider-arromichina-gw-1(config-line)#login
provider-arromichina-gw-1(config-line)#exit
provider-arromichina-gw-1(config)#line console 0
provider-arromichina-gw-1(config-line)#password cisco
provider-arromichina-gw-1(config-line)#login
provider-arromichina-gw-1(config-line)#exit
provider-arromichina-gw-1(config)#enable secret cisco
provider-arromichina-gw-1(config)#service password-encryption
provider-arromichina-gw-1(config)#username admin privilege 1 secret cisco
provider-arromichina-gw-1(config)#
```

Настройка коммутатора provider-sw-1

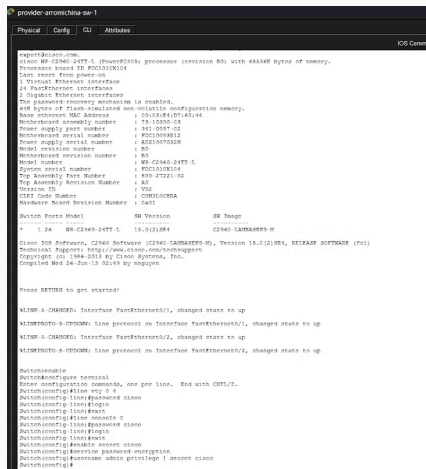


Рис. 1: Первоначально настроен коммутатор provider-sw-1

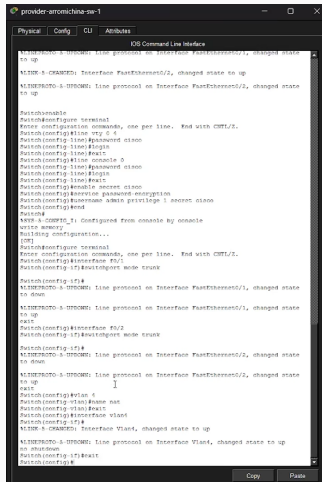
- Настроим интерфейсы маршрутизатора provider-gw-1

```
provider-arremichina-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
provider-arremichina-gw-1(config)#interface f0/0
provider-arremichina-gw-1(config-if)#no shutdown
provider-arremichina-gw-1(config-if)#exit
provider-arremichina-gw-1(config)#interface f0/0.4
provider-arremichina-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.4,
changed state to up
encapsulation dot1Q 4
provider-arremichina-gw-1(config-subif)#ip address 198.51.100.1
255.255.255.240
provider-arremichina-gw-1(config-subif)#description mka-donakaya
provider-arremichina-gw-1(config-subif)#exit
provider-arremichina-gw-1(config)#interface f0/1
provider-arremichina-gw-1(config-if)#no shut
```

Интерфейсы коммутатора

- Настроим интерфейсы коммутатора provider-sw-1



```
provider-aromichina-sw-1
Physical Config CU Attributes
IOS Command Line Interface
MINTERACTO-5-INFO: Line protocol on Interface FastEthernet0/1, changed state
to up
MINTERACTO-5-CHANGED: Interface FastEthernet0/2, changed state to up
MINTERACTO-5-INFO: Line protocol on Interface FastEthernet0/2, changed state
to up
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
Switch(config)#line vty 0 4
Switch(config-line)#password class
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#line console 0
Switch(config-line)#password class
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#enable secret class
Switch(config)#service password-encryption
Switch(config)#username admin privilege 1 secret class
Switch(config)#end
Switch#
MINTERACTO-5-INFO: Configured from console by console
with memory
Building configurations...
[OK]
Switch#configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
Switch(config)#interface f0/1
Switch(config-if)#switchport mode trunk

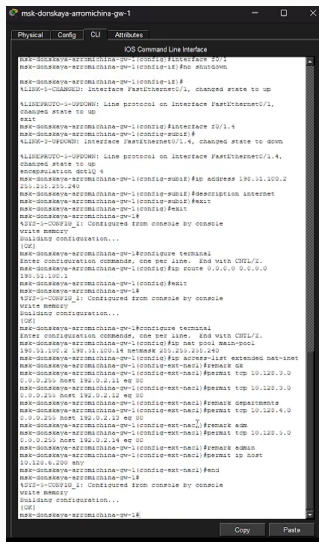
Switch(config-if)#
MINTERACTO-5-INFO: Line protocol on Interface FastEthernet0/1, changed state
to down
MINTERACTO-5-INFO: Line protocol on Interface FastEthernet0/1, changed state
to up
exit
Switch(config)#interface f0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
MINTERACTO-5-INFO: Line protocol on Interface FastEthernet0/2, changed state
to down
MINTERACTO-5-INFO: Line protocol on Interface FastEthernet0/2, changed state
to up
exit
Switch(config)#vlan 4
Switch(config-vlan)#name nas
Switch(config-vlan)#exit
Switch(config)#interface vlan4
Switch(config-if)#
MINTERACTO-5-CHANGED: Interface Vlan4, changed state to up
MINTERACTO-5-INFO: Line protocol on Interface Vlan4, changed state to up
on shutdown
Switch(config-if)#exit
Switch(config)#
```

Интерфейсы маршрутизатора



Рис. 2: Настроим интерфейсы маршрутизатора msk-donskaya-gw-1



```
mak-donskaya-atomichina-gw-1
Physical Config CU Attributes
IOS Command Line Interface
mak-donskaya-atomichina-gw-1(config)#interface f0/1
mak-donskaya-atomichina-gw-1(config-if)#no shutdown
mak-donskaya-atomichina-gw-1(config-if)#
%LINK-3-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-3-UPDOWN: Line protocol on interface FastEthernet0/1,
changed state to up
exit
mak-donskaya-atomichina-gw-1(config)#interface f0/1.4
mak-donskaya-atomichina-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.4, changed state to down
%LINEPROTO-3-UPDOWN: Line protocol on interface FastEthernet0/1.4,
changed state to up
%COMPRESSION-2-0:
mak-donskaya-atomichina-gw-1(config-subif)#ip address 192.168.100.2
255.255.255.240
mak-donskaya-atomichina-gw-1(config-subif)#description internet
mak-donskaya-atomichina-gw-1(config-subif)#exit
mak-donskaya-atomichina-gw-1(config)#exit
mak-donskaya-atomichina-gw-1#
VTYD-3-CUVDIO_1: Configured from console by console
write memory
Building configuration...
[OK]
mak-donskaya-atomichina-gw-1#configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
mak-donskaya-atomichina-gw-1(config)#ip route 0.0.0.0 0.0.0.0
192.168.100.1
mak-donskaya-atomichina-gw-1(config)#exit
mak-donskaya-atomichina-gw-1#
VTYD-3-CUVDIO_1: Configured from console by console
write memory
Building configuration...
[OK]
mak-donskaya-atomichina-gw-1#configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
mak-donskaya-atomichina-gw-1(config)#ip nat pool main-pool
192.168.100.2 192.168.100.14 netmask 255.255.255.240
mak-donskaya-atomichina-gw-1(config)#ip access-list extended nat-intet
mak-donskaya-atomichina-gw-1(config-ext-nacl)#remark GE
mak-donskaya-atomichina-gw-1(config-ext-nacl)#permit tcp 10.120.3.0
0.0.0.255 host 192.0.2.11 eq 80
mak-donskaya-atomichina-gw-1(config-ext-nacl)#permit tcp 10.120.3.0
0.0.0.255 host 192.0.2.12 eq 80
mak-donskaya-atomichina-gw-1(config-ext-nacl)#remark departmente
mak-donskaya-atomichina-gw-1(config-ext-nacl)#permit tcp 10.120.4.0
0.0.0.255 host 192.0.2.13 eq 80
mak-donskaya-atomichina-gw-1(config-ext-nacl)#remark adm
mak-donskaya-atomichina-gw-1(config-ext-nacl)#permit tcp 10.120.3.0
0.0.0.255 host 192.0.2.14 eq 80
mak-donskaya-atomichina-gw-1(config-ext-nacl)#remark admin
mak-donskaya-atomichina-gw-1(config-ext-nacl)#permit ip host
19.120.6.200 any
mak-donskaya-atomichina-gw-1(config-ext-nacl)#end
mak-donskaya-atomichina-gw-1#
VTYD-3-CUVDIO_1: Configured from console by console
write memory
Building configuration...
[OK]
mak-donskaya-atomichina-gw-1#
```

```
mak-donskaya-arr@omichina-qw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
mak-donskaya-arr@omichina-qw-1(config)#ip nat inside source list nat-
inet pool MAIN-pool overload
mak-donskaya-arr@omichina-qw-1(config)#int 10/0.3
mak-donskaya-arr@omichina-qw-1(config-subif)#ip nat inside
mak-donskaya-arr@omichina-qw-1(config-subif)#exit
mak-donskaya-arr@omichina-qw-1(config)#interface 10/0.101
mak-donskaya-arr@omichina-qw-1(config-subif)#ip nat inside
mak-donskaya-arr@omichina-qw-1(config-subif)#exit
mak-donskaya-arr@omichina-qw-1(config)#interface 10/0.102
mak-donskaya-arr@omichina-qw-1(config-subif)#ip nat inside
mak-donskaya-arr@omichina-qw-1(config-subif)#exit
mak-donskaya-arr@omichina-qw-1(config)#interface 10/0.103
mak-donskaya-arr@omichina-qw-1(config-subif)#ip nat inside
mak-donskaya-arr@omichina-qw-1(config-subif)#exit
mak-donskaya-arr@omichina-qw-1(config)#interface 10/0.104
mak-donskaya-arr@omichina-qw-1(config-subif)#ip nat inside
mak-donskaya-arr@omichina-qw-1(config-subif)#exit
mak-donskaya-arr@omichina-qw-1(config)#interface 10/1.4
mak-donskaya-arr@omichina-qw-1(config-subif)#ip nat outside
mak-donskaya-arr@omichina-qw-1(config-subif)#exit
mak-donskaya-arr@omichina-qw-1(config)#
```

Рис. 4: Настроим NAT

Настройка доступа

```
RRR-gonskaya-aironachana-gw-1(config)#ip nat inside source static tcp
10.120.0.2 80 190.51.100.2 80
RRR-gonskaya-aironachana-gw-1(config)#ip nat inside source static tcp
10.120.0.3 20 190.51.100.3 20
RRR-gonskaya-aironachana-gw-1(config)#ip nat inside source static tcp
10.120.0.3 21 190.51.100.3 21
RRR-gonskaya-aironachana-gw-1(config)#ip nat inside source static tcp
10.120.0.4 25 190.51.100.4 25
RRR-gonskaya-aironachana-gw-1(config)#ip nat inside source static tcp
10.120.0.4 110 190.51.100.4 110
RRR-gonskaya-aironachana-gw-1(config)#ip nat inside source static tcp
10.120.0.200 3309 190.51.100.10 3309
RRR-gonskaya-aironachana-gw-1(config)#end
RRR-gonskaya-aironachana-gw-1#
RRR-gw-1-CONF10-1: Configured from console by console
Write Memory
Building configuration...
[OK]
RRR-gonskaya-aironachana-gw-1#
```

Рис. 5: Настроим доступа из Интернета

Вывод

- Мы смогли приобрести практические навыки по настройке доступа локальной сети к внешней сети посредством NAT